

Tuning the Kondo temperature of MoS₂ mirror twin boundaries on graphene

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Mirror twin boundaries (MTBs) are one-dimensional line defects that exhibit a Kondo effect [1,2]. It was shown that this effect arises from confining a spin $\frac{1}{2}$ along the boundary, which is screened by the bath electrons. The dI/dV signatures of the Kondo effect in monolayer MoS₂ MTBs are the zero-bias resonance peak, in addition to a singly occupied state below the Fermi level E_F and a doubly unoccupied state above E_F [1]. The experimental data give access to the Anderson model parameters: the splitting U of the non-degenerate states, the impurity orbital's position ε with respect to E_F , and the impurity level's bare width γ_0 . In this work, we compare the Kondo temperature (T_K) measured in MTBs of MoS₂ grown on three different substrates. In the case of MoS₂ / Graphene (Gr) / Ir(111), T_K was in the range $[10^{-4} \text{ K}, 10^{-10} \text{ K}]$ [1]. By intercalating a single layer of Eu under Gr, we noticed a decrease in γ_0 , leading to a drop in T_K . In the third system, MoS₂ / Gr / Ir(110), T_K increased significantly to the range $[1 \text{ K}, 10 \text{ K}]$. This enabled us to study the Kondo effect at the boundary between the weakly screened magnetic state above T_K and the Kondo singlet state below T_K .

[1] Camiel van Efferen, Jeison Fischer, Theo A. Costi, Achim Rosch, Thomas Michely and Wouter Jolie, *Nat. Phys* **20**, 82-87 (2024).

[2] Mahasweta Bagchi, Tfyeché Y. Tounsi, Affan Safeer, Camiel van Efferen, Achim Rosch, Thomas Michely, Wouter Jolie, Theo A. Costi, and Jeison Fischer, *Phys. Rev. Lett* **133**, 246701 (2024).